

Squishy Circuits - Information for Lead

Squishy Circuits - Game Description

Students will create circuits using play-doh & LED's to solve logic problems.

Resource List

In the room you should have:

Students will get:

- *Tubs of play-doh (1 tub per group)*
- *9v Batteries (1 per group)*
- *Snap-on wires connectors (1 per group)*
- *10mm LED lights (1 per group + spares)*
- *A3 paper or butcher's paper (1 sheet per group)*
- *Pens or markers (around 2 per group is preferable)*
- *Squishy Circuits student handouts (1 per group)*
- *Squishy Circuits Bonus Questions student handouts (1 per group)*

Other:

- *Tutor answers handout*
- *Computerless rewards (Freddos/Mentos)*

Setup

Resources

There are no posters to put up for this worksheet. Materials should be at the front of the room to be handed out.

Tutors

Tutors should help handout the materials to the groups (particularly the batteries, connectors, and LED's)

They will have an answer sheet they can use to help groups, and check answers. A tutor should check the answers to the basic question sheet, before giving the students a reward + the bonus questions handout. You may want to have a couple of tutors stationed at the front of the room for this. The rest can be floating around helping groups as needed.

Students

Students should stay in one area, preferably in groups of 4-6.

The playdough should be kept on a piece of butchers/A3 paper to protect the surfaces students are working on.

Each group should have just 1 colour of play-doh to avoid muddying the colours.

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Running the Game

This game has up to 3 sections

1. Squishy Circuit basic questions - reward students for completing
2. Squishy Circuit bonus questions [OPTIONAL]
3. Free Design competition [OPTIONAL]

You may want to run a timer on the screen

If you have extra time after the bonus questions, have students come up with their own designs/circuits. You may want to make it a competition, announcing the top 3 most creative designs.

Troubleshooting

If the LED's are not lighting up

- Try making the playdough 'wires' thinner
- Try switching the direction of the LED
- Try the circuit with an LED or battery that is confirmed to work

Finishing the Game

To wrap this game up you may want to recap the types of logic gates they have created, and how they can be useful for when you have a repeatable scenario (such as checking the weather every day)

If you run a design competition you may want to show pictures of the winning designs on the board, or congratulate the teams before asking students to pack-up.